Web Dev Mastery

📘 CSS Units

✅ What We Cover:

- Absolute Units
    - `px`, `pt`, `cm`, `mm`, `in`, `pc`
- Relative Units
    - `em`, `rem`, `%`, `vw`, `vh`, `vmin`, `vmax`, `ch`, `ex`
- Best Practices: When to Use Which Unit
- Code Examples for Each Unit

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1. 📏 Absolute Units

Absolute units are fixed and **do not scale** according to the screen size or parent elements.

Unit	Meaning	Description
px	Pixels	Fixed-size dot on the screen. Most common unit.
pt	Points	1pt = 1/72 inch (used in print)
cm	Centimeters	Used for print media
mm	Millimeters	Used for print media

in	Inches	1in = 2.54 cm
pc	Picas	1pc = 12pt

Example:

```
.box {
  width: 200px;
  height: 100px;
  font-size: 12pt;
  border: 0.1cm solid black;
}
```

🟢 **Use When:** You need fixed dimensions like in icons, layout containers, or print styles.

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2. 🔁 Relative Units

Relative units depend on **parent elements, root elements**, or **viewport size**. They are **responsive** and ideal for dynamic designs.

2.1 `em` — Relative to the parent element's font size

- `1em` = current element's font size
- Nested `em` multiplies

```
.container {
  font-size: 16px;
}
.child {
  font-size: 1.5em; /* 24px */
}
```

🟢 **Use For:** Padding, margin, spacing based on text size

2.2 `rem` — Relative to the root element's font size (`html`)

- Avoids nesting problems of `em`
- `1rem` = font-size of `<html>` (commonly 16px)

```
html {
  font-size: 16px;
}
p {
  font-size: 1.25rem; /* 20px */
}
```

🟢 **Use For:** Font sizes, spacing for consistency across components

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2.3 `%` — Relative to the parent container's dimensions

```
.parent {
  width: 500px;
}
.child {
  width: 50%; /* 250px */
}
```

🟢 **Use For:** Widths, heights, font sizes (sometimes)

2.4 `vw` / `vh` — Relative to viewport width and height

- `1vw` = 1% of viewport width
- `1vh` = 1% of viewport height

```
.hero {
  width: 100vw;
  height: 100vh;
}
```

🟢 **Use For:** Full-screen sections, responsive layouts

2.5 `vmin` / `vmax`

- `vmin`: 1% of the smaller dimension (width or height)
- `vmax`: 1% of the larger dimension

```
.section {
  font-size: 3vmin;
}
```

🟢 **Use For:** Typography that scales with screen orientation

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2.6 `ch` — Width of the character `0`

```
input {
  width: 30ch; /* approx 30 characters wide */
}
```

🟢 **Use For:** Inputs and text areas that fit a specific number of characters

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2.7 `ex` — x-height of the current font

- Rarely used
- Based on height of lowercase `x` in font

✅ Summary: When to Use Which Unit

Use Case	Recommended Units
Font Sizes	`rem`, `em`
Spacing (margin/padding)	`em`, `rem`
Container Widths	`%`, `vw`, `rem`
Full Page Sections	`vh`, `vw`
Character-limited Inputs	`ch`
Fixed-size Elements	`px`
Print Styles	`cm`, `mm`, `pt`

🧪 Combined Example

```
html {
  font-size: 16px;
}

body {
  margin: 0;
  padding: 2rem;
}

.container {
  width: 80vw;
  height: 100vh;
  font-size: 1rem;
  padding: 1em;
}

.button {
  font-size: 1.2rem;
  padding: 0.5em 2em;
  margin-top: 2%;
}
```

✅ Best Practices

- Prefer `rem` for **scalable typography**

- Use `em` when you want size based on **parent**
- Use `%`, `vw`, `vh` for **responsive layouts**
- Avoid `px` for fonts if aiming for accessibility
- Use `ch` for character-based input sizing